

CRIME AND SOCIOECONOMIC FACTORS IN CHICAGO: A DATA-DRIVEN APPROACH

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1 MOTIVATION

Crime impacts public safety and community well-being, but many models overlook key factors. Understanding how income, housing, and employment influence crime can lead to better prevention strategies. Our approach integrates crime data with socioeconomic indicators to provide deeper insights and guide effective policy decisions.

3 OUR APPROACH

Our approach combines machine learning and interactive data visualization to identify influential social determinants driving crime patterns in Chicago. By analyzing a broad set of variables, we aim to determine which predictors have the greatest impact on crime rates.

Two-Step Variable Selection

1. Lasso Regression:

Lasso imposes L1 penalty on the estimated coefficients, forcing those with less impact on crime rate to zero, and effectively "selecting" features with non-zero coefficients. The selected features were then cross-validated using a Ridge model, which imposes L2 penalty.

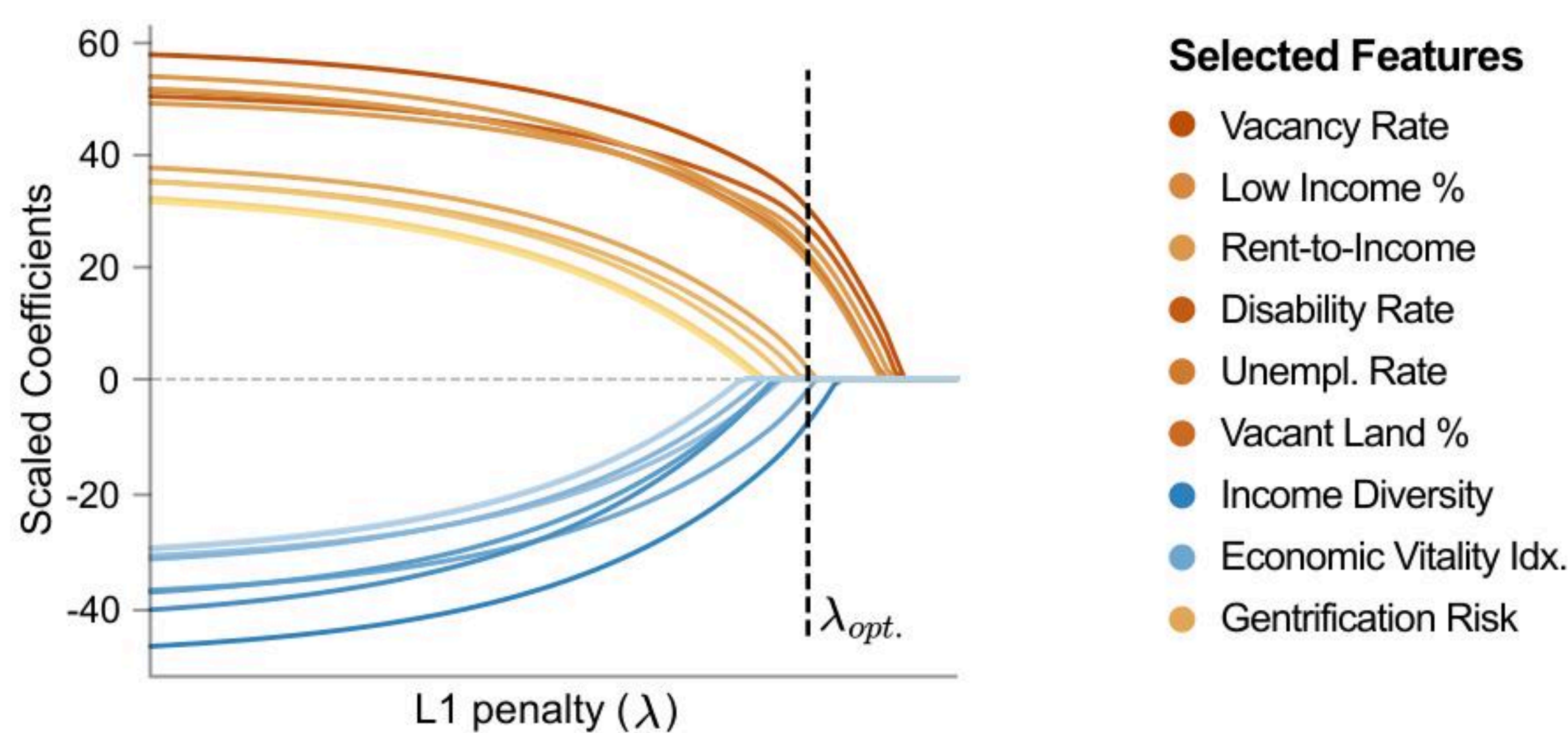
$$\arg \min_{\beta} \left(L(\beta, \mathbf{X}, \mathbf{y}) + \lambda_1 \|\beta\|_1 + \lambda_2 \sum \|\beta_{\text{group}}\|_2 \right)$$

Lasso Equation

2. Eliminating Highly Correlated Features:

Highly correlated selected features were examined. For each pair, the feature with lower Lasso coefficient was eliminated.

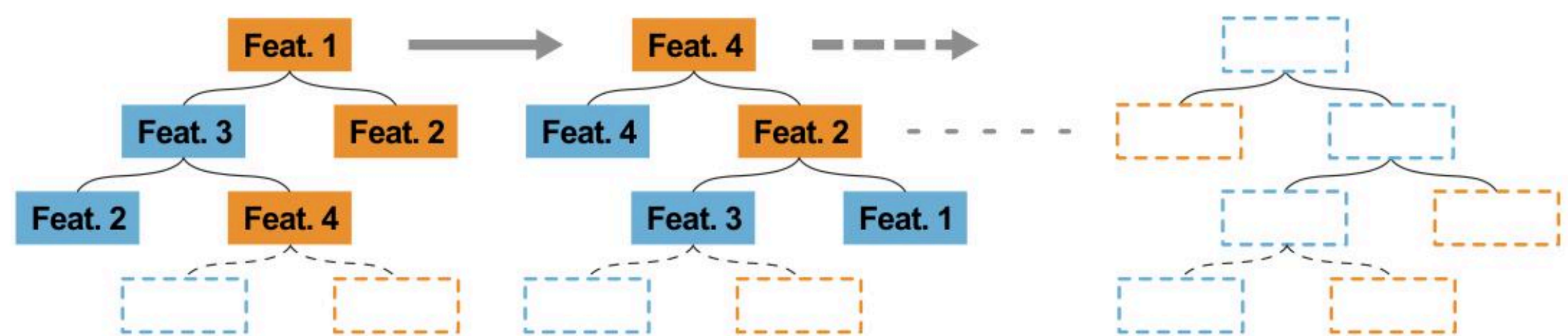
After 2 iterations, our model selected 9 features



Lasso Paths: 9 out of 19 features selected

Capturing Non-Linear Relationships with XGBoost

The final set of features were validated using an XGBoost model, which is a tree-based ensemble model that can capture non-linear patterns in the data. Feature were ranked based on their "Importance" score obtained from the model.



Ensemble Trees: New trees correct errors of previous trees
Tree architecture helps capture non-linear patterns

Why is this effective?

- Crime is influenced by multiple, interdependent factors.
- XGBoost handles nonlinear relationships and selects the strongest predictors from high-dimensional data.
- LASSO improves interpretability and prevents overfitting through regularization, ensuring robust predictions..

2 DATA

We analyze crime data from the City of Chicago Data Portal, covering over 2.5 million crime records aggregated across 77 community areas. Additionally, we incorporate 250+ socioeconomic indicators from the Community Snapshot Data, refining them into 86 engineered features. These include economic factors, housing stability, and education levels, providing a comprehensive view of crime patterns.

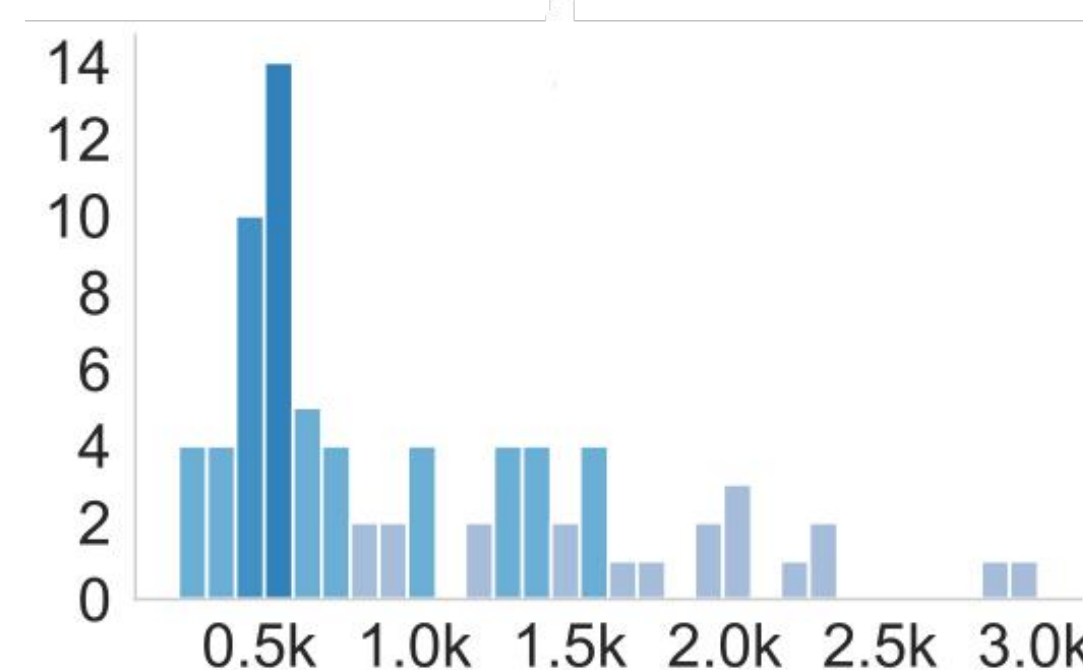


CHICAGO
DATA PORTAL



Chicago Metropolitan
Agency for Planning

Distribution of Crime Rate for Each Community Area



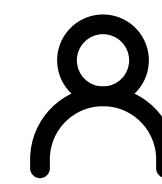
2.5 MILLION CRIME RECORDS

250+ INDICATORS THAT CAN BE CATEGORIZED INTO:

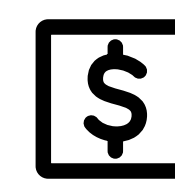
TRANSPORTATION



DEMOGRAPHIC



SOCIOECONOMIC



DISABILITY



EDUCATION



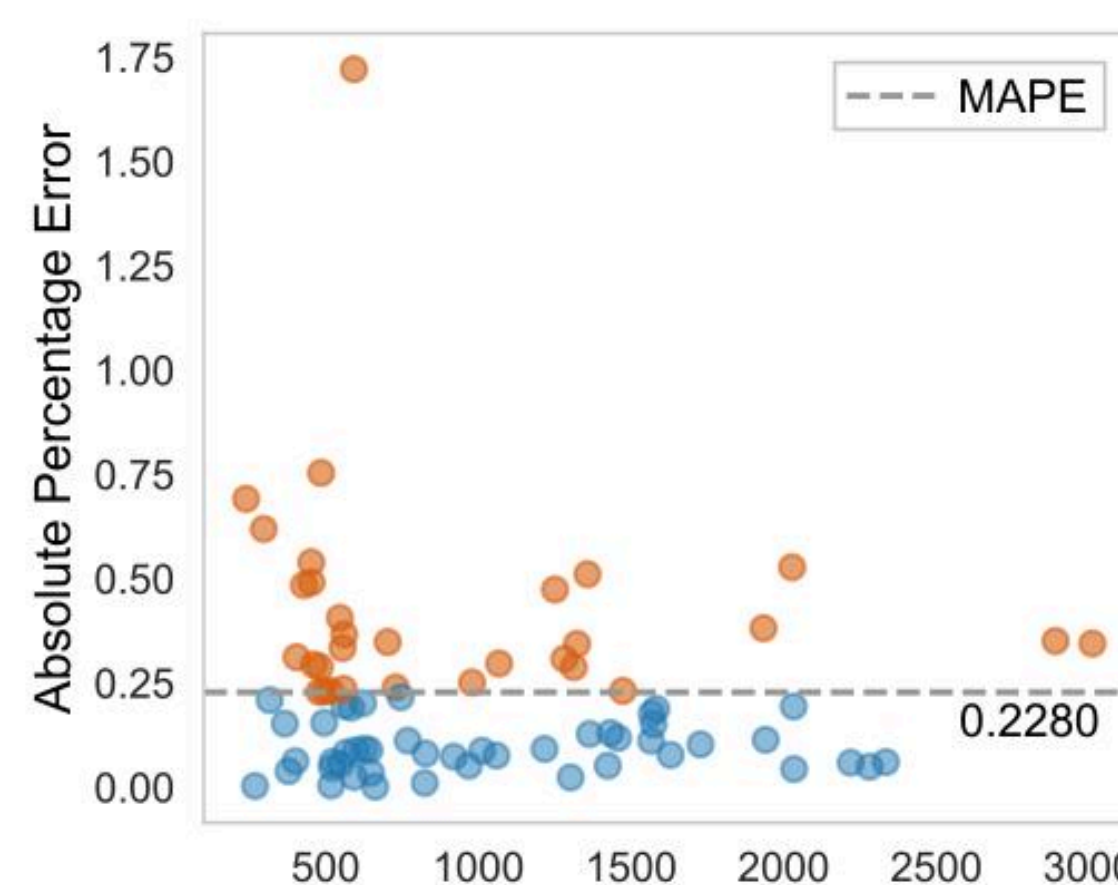
HOUSING



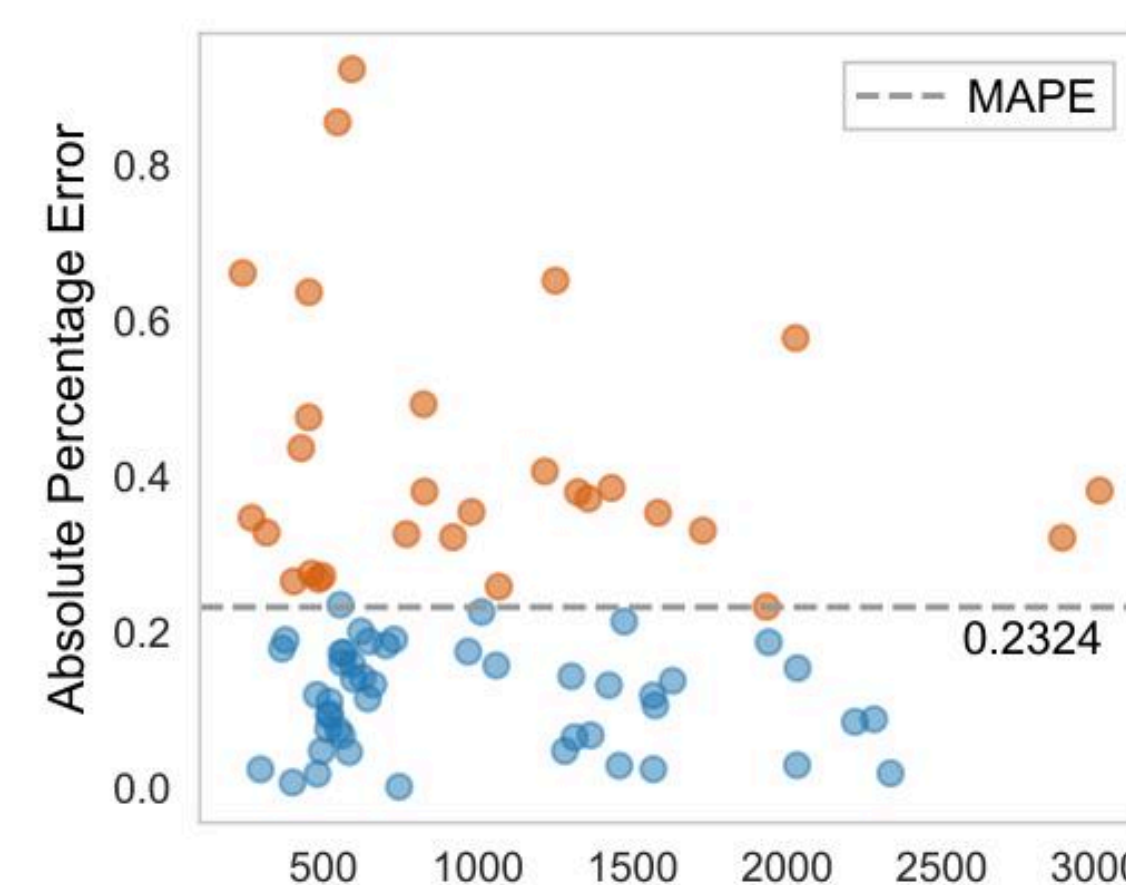
4 EVALUATION AND RESULTS

Evaluation

Our approach was validated using cross-validation and residual analysis, comparing an initial model with 86 features to a refined model with fewer, more relevant features. While the average error (MAPE) remained similar, the highest residual error dropped significantly (175% → 90% MAPE), improving prediction reliability.



Full Feature Set



Final 9 Feature Set

Absolute Percentage Error vs. Crime Rate per Community Area for baseline and final XGBoost models

Results

Our results highlight key socioeconomic factors influencing crime rates below. Compared to traditional models relying solely on historical crime data, our approach integrates broader socioeconomic indicators, leading to a more holistic and interpretable understanding of crime patterns. This enables better policy recommendations and resource allocation.

Final Ranking of Features

Feature	Score
Unemployment Rate	0.2154
Rent-to-Income	0.2086
Economic Vitality Index	0.1938
Vacancy Rate	0.1938
Disability Rate	0.0712